6 **(a)** **(i)** Σ Rest masses before $= 2 \times 3.34454 \times 10^{-27}$ kg $= 6.68908 \times 10^{-27}$ kg
Σ Rest masses after $= 5.00835 \times 10^{-27} + 1.67496 \times 10^{-27}$
$= 6.68331 \times 10^{-27}$ kg

Difference in rest masses, $\Delta m = 5.77 \times 10^{-30}$ kg [1]

Using $\Delta E = \Delta mc^2$, Energy released $= 5.77 \times 10^{-30} \times c^2$
$= 5.193 \times 10^{-13}$ J
$= 5.19 \times 10^{-13}$ J [1]

Note: 1m for answer, 1m for method. Give 1m if student gave a negative (but correct value)

(ii) Since the total energy released is larger than calculated, the reactants must have possessed kinetic energy before the reaction. [1]

(b) **(i)** Loss in KE = Gain in EPE

$$2K - 0 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{2r} - 0$$ [1]

$$K = \frac{1}{16\pi\varepsilon_0}\frac{e^2}{2r} = \frac{1}{16\pi\varepsilon_0}\frac{(1.6 \times 10^{-19})^2}{(1 \times 10^{-15})} = 5.75 \times 10^{-14} = 5.8 \times 10^{-14} \text{ J}$$ [1]

Note: Accept if student gives 3 sf (Being lenient)

(ii) $K = \frac{3}{2}kT$

$T = 2.78 \times 10^9 = 2.8 \times 10^9$ K [1, ECF]

Note: Accept if student gives 3 sf (Being lenient)

(c) **(i)** The total energy required to separate a nucleus (completely) into its constituent particles. [1]

Accept: Total energy release when totally separated nucleons come together to form a nucleus.

(ii) X is the **<u>neutron</u>**. Because it is a **<u>free / individual / single particle</u>** here {or because it is not within a nucleus} it has no binding energy. [1]

(iii) Binding Energy of Beryllium = (6.4763)(9) = 58.2867 MeV
Binding Energy of Helium = (7.0819)(4) = 28.3276 MeV
Total Binding Energy of Reactants = 58.2867 + 28.3276 = 86.6143 MeV
Binding Energy of Carbon = (7.6885)(12) = 92.262 MeV
[1 for all 4 values]

Net energy released in the process = B.E(products) - B.E(reactants)
= 92.262 - 86.6143
= 5.65 MeV [1]
$= 5.65 \times 10^6 \times 1.6 \times 10^{-19}$ J
$= 9.04 \times 10^{-13}$ J [1]

Note: Full ecf for 2nd and 3rd mark.
Give 1 m for each correct method (i.e. BE(pdt) – BE(rt), convert MeV to J)